ii. the central metal ion/atom is coordinated to more than one kind of ligands is called a heteroleptic complex. Example, $[Co(NH_3)_5Cl]^{2+}$, $[Pt(NH_3)_2Cl_2)]$

5.4 Nomenclature of coordination compounds

In the earlier days, the compounds were named after their discoverers. For example, $K[PtCl_3(C_2H_4)]$ was called Zeise's salt and $[Pt(NH_3)_4][PtCl_4]$ is called Magnus's green salt etc... There are numerous coordination compounds that have been synthesised and characterised. The International Union of Pure and Applied Chemistry (IUPAC) has developed an elaborate system of nomenclature to name them systematically. The guidelines for naming coordination compounds based on IUPAC recommendations (2005) are as follows:

1. The cation is named first, followed by the anion regardless of whether the ion is simple or complex. For example
   - In $K_4[Fe(CN)_6]$, the cation K^+ is named first followed by$[Fe(CN)_6]^{4-}$.
   - In $[Co(NH_3)_6]Cl_3$, the complex cation $[Co(NH_3)_6]^{3+}$ is named first followed by the anion Cl^-
   - In $[Pt(NH_3)_4][PtCl_4]$, the complex cation $[Pt(NH_3)_4]^{2+}$is named first followed by the complex anion $[PtCl_4]^{2-}$

2. The simple ions are named as in other ionic compounds. For example,

Simple cation	Symbol	Simple anion	Symbol
Sodium	Na^+	Chloride	Cl^-
Potassium	K^+	Nitrate	NO_3^-
Copper	Cu^{2+}	Sulphate	SO_4^{2-}

3. To name a complex ion, the ligands are named first followed by the central metal atom/ion. When a complex ion contains more than one kind of ligands they are named in alphabetical order.

   a. Naming the ligands:

   i. The name of anionic ligands ends with the letter 'o' and the cationic ligand ends with 'ium'. The neutral ligands are usually called with their molecular names with fewer exceptions namely, H_2O (aqua), CO (carbonyl), NH_3 (ammine) and NO (nitrosyl).

   ii. A κ-term is used to denote an ambidendate ligand in which more than one coordination mode is possible. For example, the ligand thiocyanate can bind to the central atom/ion, through either the sulfur or the nitrogen atom. In this ligand, if sulphur forms a coordination bond with metal then the ligand is named thiocyanato-κS and if nitrogen is involved, then it is named thiocyanato-κN.

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